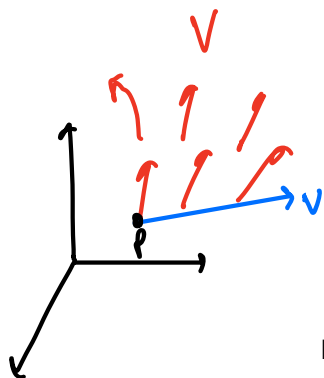


Differential Geometry 5 Sep

Last time:



Brutal

covariant derivative
of a vector field

$$\left(\nabla_v V \right)_p = \frac{d}{dt} \Big|_{t=0} V(p+tv).$$

↑
tangent vector at p.

unity:<https://aluminum1.github.io/unityapps/Ep8web>

" ω_{ij} is the 1-form which, when evaluated on a tangent vector $V \in T_p \mathbb{R}^3$, tells you the j th component of how E_i varies in the direction of V

→ it's a matrix of 1-forms,

$$\omega = \begin{pmatrix} \omega_{11} & \omega_{12} & \omega_{13} \\ \omega_{21} & \omega_{22} & \omega_{23} \\ \omega_{31} & \omega_{32} & \omega_{33} \end{pmatrix}$$

Theorem $\omega = dA A^T$

↑ Exercise

$$\begin{aligned} E_1 &= a_{11} U_1 + a_{12} U_2 + a_{13} U_3 \\ E_2 &= a_{21} U_1 + a_{22} U_2 + a_{23} U_3 \end{aligned}$$

$$A_{ij} = E_i \cdot U_j$$

$$\begin{aligned}
 \omega_{ij} &= \sum_k \omega_{ij}(U_k) dx_k; \quad \omega_{ij}(U_k) = \nabla_{U_k} E_i \cdot E_j \\
 &= -E_j \cdot \nabla_{U_k} E_i \\
 &= -E_j \cdot \nabla_{U_k} \left[\sum_p a_{ip} U_p \right] \\
 &= -E_j \cdot \sum_p \left(\frac{\partial a_{ip}}{\partial x_k} U_p + 0 \right)
 \end{aligned}$$

$$\begin{aligned}
 \therefore \omega_{ij} &= - \sum_{p,k} \frac{\partial a_{ip}}{\partial x_k} \underbrace{E_j \cdot U_p}_{a_{jp}} dx_k \\
 &= - \sum_p (dA)_{ip} A_{pj}^T
 \end{aligned}$$

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Proof

$$\begin{aligned}
 \theta_i &= \sum_j \theta_i(U_j) dx_j \\
 &= \sum_{j,k} A_{kj} \underbrace{\theta_i(E_k)}_{\delta_{ik}} dx_j
 \end{aligned}$$

$$\text{But, } U_j = \sum_k A_{kj} E_k$$

$$= \sum_j A_{ij} dx_j$$

$$\therefore \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \\ A_{31} & A_{32} \end{bmatrix} \begin{bmatrix} dx_1 \\ dx_2 \\ dx_3 \end{bmatrix}$$

$$\therefore d\theta = d(A dx)$$

$$= dA \wedge dx + A \underbrace{d^2 x}_{=0}$$

$$= dA A^T A \wedge dx$$

$$= \underbrace{dA A^T}_{=\omega} \wedge \underbrace{A dx}_{\theta}$$

$$d(f\omega) = df \wedge \omega + f d\omega$$

$$\begin{aligned} (2) \quad d\omega &= d(dA A^T) \\ &= -dA \wedge dA^T \\ &= -\omega A \wedge A^T \omega^T \\ &= -\omega \wedge \omega^T \\ &= \omega \wedge \omega \end{aligned}$$

$$\omega = dA A^T$$

$$\boxed{\omega^T = A dA^T}$$

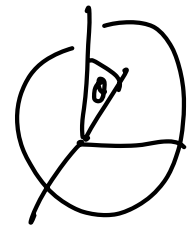
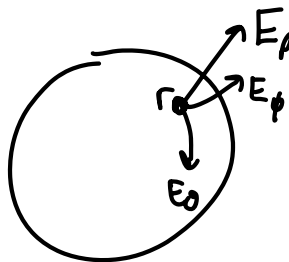
$$\therefore dA^T = A^T \omega^T$$

$$\text{i.e. } dA = \omega A$$

$$\begin{aligned} d(df \cdot g) &= d(g df) \\ &= dg \wedge df \\ &= -df \wedge dg \end{aligned}$$

eg. Spherical frame field

$$\theta = A dx$$



$$\begin{aligned} x &= r \sin\theta \cos\phi \\ y &= r \sin\theta \sin\phi \\ z &= r \cos\theta \end{aligned}$$

$$\Rightarrow \begin{aligned} E_r &= \text{normalize} \left(\frac{\partial \rho}{\partial r} \right) = \frac{xU_1 + yU_2 + zU_3}{r} \\ E_\theta &= \text{normalization of } \frac{\partial \rho}{\partial \theta} \end{aligned}$$

$$A =$$

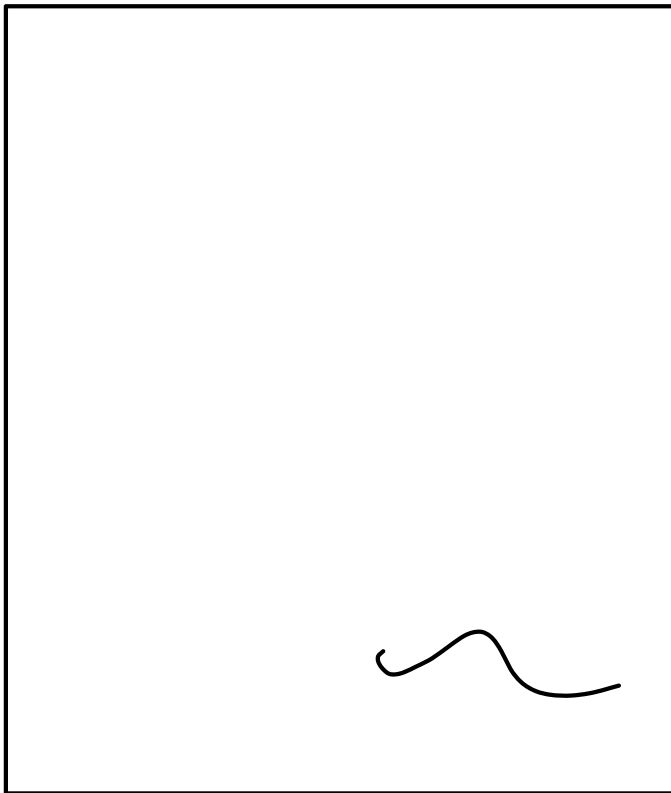
If E_i is the frame field of a coordinate system $p = p(\theta)$

$$\theta = A dx.$$

$$\text{Claim } \theta_a = \sum_i \frac{\partial p}{\partial x_i} dx_i$$

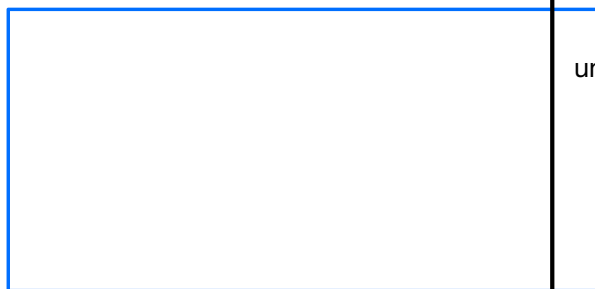
$$A = \begin{bmatrix} E_i \cdot U_j \end{bmatrix}$$

Lesson



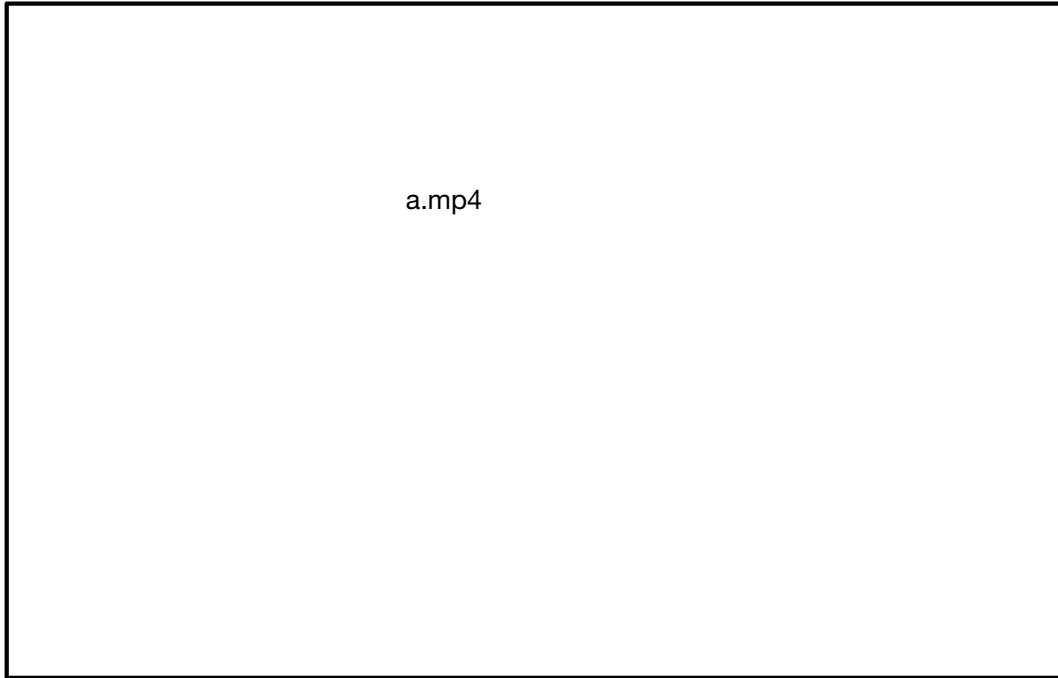
Example

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unity:<https://aluminum1.github.io/unityapps/Ep8web>

And here's some writing

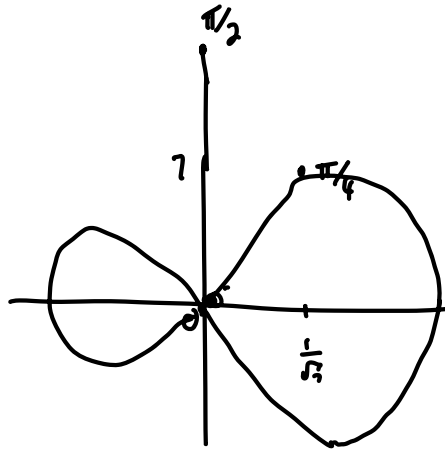


And save now.

4.1 Surfaces

Defn A coordinate patch $\alpha : D \rightarrow \mathbb{R}^3$ is an injective immersion of an open set $D \subseteq \mathbb{R}^2$ into $\mathbb{R}^3$.

Defn A subset $M \subseteq \mathbb{R}^3$ is called a surface if for each $p \in M$ there exists



$(\sin t, \sin 2t)$

$\pi/4$	$-\frac{1}{\sqrt{2}}$	1
$\frac{\pi}{2}$	1	0
$\frac{3\pi}{4}$	$-\frac{1}{\sqrt{2}}$	-1
π		